Practical 17: SQLite CRUD Operations

Source: MODULE 2.pdf | Source pages: 55–64

Aim

To develop and test an Android application demonstrating sqlite crud operations.

Steps to Perform the Practical

1. Create an Android Studio project for the SQLite CRUD practical.

2. Design the layout with the input fields and buttons shown in the source code.

3. Create a SQLiteOpenHelper class and define the database name, table and columns.

4. Implement insert, read, update and delete operations using SQLiteDatabase.

5. Connect the buttons in MainActivity.java to the corresponding database methods.

6. Run the app. Insert a record, view it, update it, delete it, and verify each result.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

Database and Networking

10. Develop an Android application implementing SQLite database CRUD operations.

ProjectName: SQLLiteCRUDDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <EditText

       android:id="@+id/etName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter Name"/>

   <EditText

       android:id="@+id/etNewName"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:hint="Enter New Name"/>

   <Button

       android:id="@+id/btnInsert"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"
```

```
        android:text="Insert"/>

    <Button

        android:id="@+id/btnView"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="View Data"/>

    <Button

        android:id="@+id/btnUpdate"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Update"/>

    <Button

        android:id="@+id/btnDelete"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:text="Delete"/>

    <TextView

        android:id="@+id/txtResult" android:layout_width="match_parent"

        android:layout_height="wrap_content"

        android:textSize="18sp"

         android:paddingTop="20dp"/>

</LinearLayout>
```

Create a Database Helper Class.

Right Click:

java package → New → Java Class

Class Name: DBHelper

```
package com.example.sqllitecruddemo;


import android.content.ContentValues;

import android.content.Context;

import android.database.Cursor;

import android.database.sqlite.SQLiteDatabase;

import android.database.sqlite.SQLiteOpenHelper;


public class DBHelper extends SQLiteOpenHelper {


    public DBHelper(Context context) {

        super(context, "StudentDB", null, 1);

    }


    @Override

    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE student(name TEXT)");

    }


    @Override

    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        db.execSQL("DROP TABLE IF EXISTS student");
```

```
        onCreate(db);
    }


    // Insert
    public void insertData(String name) {
        SQLiteDatabase db = this.getWritableDatabase();


        ContentValues values = new ContentValues();
        values.put("name", name);


        db.insert("student", null, values);
    }


    // View
    public String viewData() {
        SQLiteDatabase db = this.getReadableDatabase();


        Cursor cursor = db.rawQuery("SELECT * FROM student", null);


        StringBuilder data = new StringBuilder();


        while (cursor.moveToNext()) {
            data.append(cursor.getString(0)).append("\n");
        }


        cursor.close();
```

```
    return data.toString();
}

// Update
public void updateData(String oldName, String newName) {
    SQLiteDatabase db = this.getWritableDatabase();

    ContentValues values = new ContentValues();
    values.put("name", newName);

    db.update(
            "student",
            values,
            "name=?",
            new String[]{oldName}
    );
}

// Delete
public void deleteData(String name) {
    SQLiteDatabase db = this.getWritableDatabase();

    db.delete(
            "student",
            "name=?",
```

```
                new String[]{name}

        );

    }

}
```

MainActivity.java

```
package com.example.sqllitecruddemo;



import android.os.Bundle;

import android.widget.Button;

import android.widget.EditText;

import android.widget.TextView;

import android.widget.Toast;



import androidx.appcompat.app.AppCompatActivity;



public class MainActivity extends AppCompatActivity {



    EditText etName, etNewName;

    TextView txtResult;



    Button btnInsert, btnView, btnUpdate, btnDelete;



    DBHelper dbHelper;
```

```java
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    etName = findViewById(R.id.etName);
    etNewName = findViewById(R.id.etNewName);
    txtResult = findViewById(R.id.txtResult);

    btnInsert = findViewById(R.id.btnInsert);
    btnView = findViewById(R.id.btnView);
    btnUpdate = findViewById(R.id.btnUpdate);
    btnDelete = findViewById(R.id.btnDelete);

    dbHelper = new DBHelper(this);

    // Insert
    btnInsert.setOnClickListener(v -> {

        String name = etName.getText().toString();

        dbHelper.insertData(name);

        Toast.makeText(
                this,
                "Data Inserted",
```

```
                Toast.LENGTH_SHORT
        ).show();
    });

    // View
    btnView.setOnClickListener(v -> {

        String data = dbHelper.viewData();

        txtResult.setText(data);
    });

    // Update
    btnUpdate.setOnClickListener(v -> {

        String oldName = etName.getText().toString();
        String newName = etNewName.getText().toString();

        dbHelper.updateData(oldName, newName);

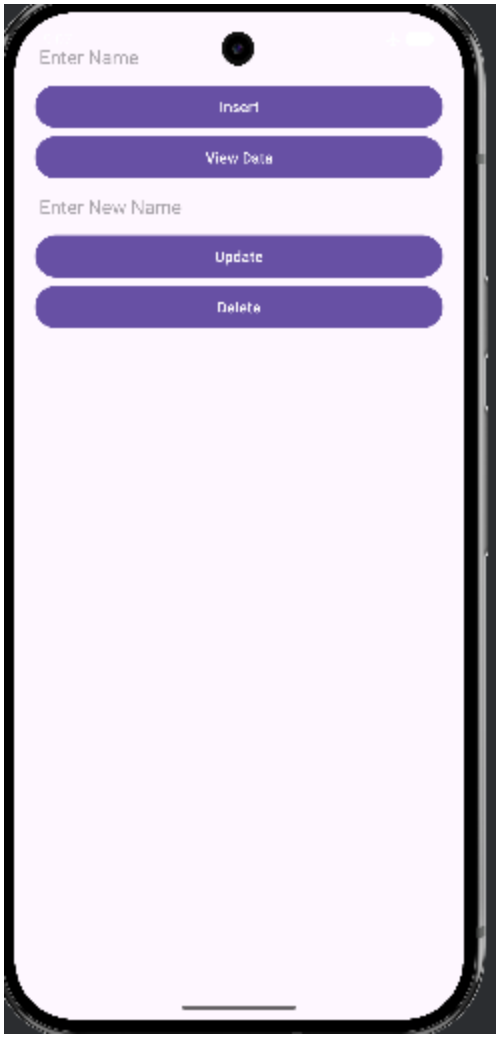
        Toast.makeText(
                this,
                "Data Updated",
                Toast.LENGTH_SHORT
        ).show();
    });
```

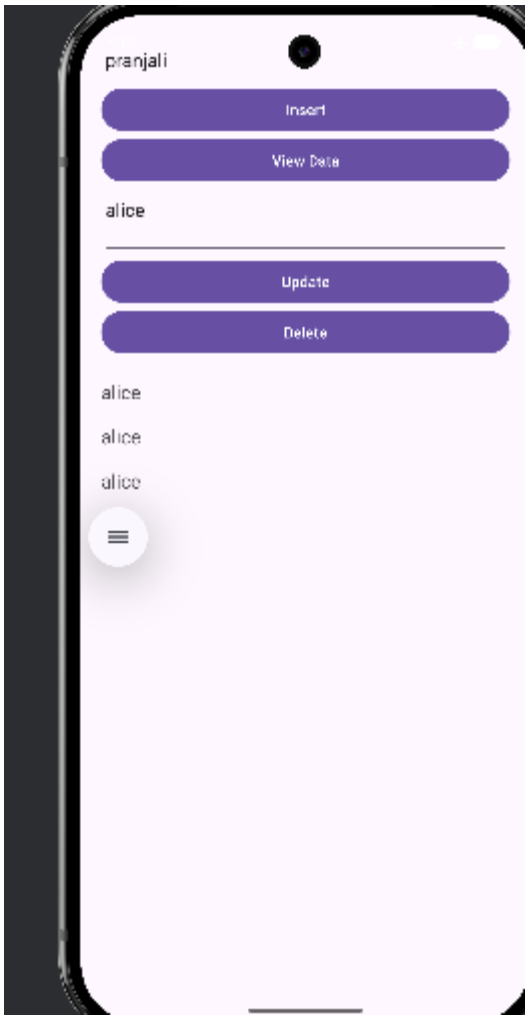
```
        // Delete
        btnDelete.setOnClickListener(v -> {

            String name = etName.getText().toString();

            dbHelper.deleteData(name);

            Toast.makeText(
                    this,
                    "Data Deleted",
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


Enter Name
Insert
View Data
Enter New Name
Update
Delete


pranjali
Insert
View Data
alice
Update
Delete
alice
alice
alice